

RISC-V-based AES Accelerator for Wearable Devices

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Arguably, the most important aspect in an interconnected world is data, with a constant stream of it flowing in and out of our phones, watches, or computers, and keeping it safe is just as important. Coinciding with the rise of wireless wearable communication (e.g. earbuds, smart watches, smart rings, etc), we have increased demand for more efficient cryptography, as these devices generally require low power consumption within a small area. As protecting data transmitted from wireless wearable devices is paramount to consumer safety, we seek to implement a hardware module to accelerate AES encryption/decryption operations as software solutions prove to be less secure or less efficient. The project focuses on fully developing a System-on-Chip (SoC) with a RISC-V CPU for general purpose processing, an AXI bus to facilitate communication between modules, an AES accelerator for more power efficient cryptography, and an SRAM for storing data and the current instruction. The SoC will be designed with power efficiency and area in mind, as wearable devices have strong constraints with battery life and size. We plan to start with an implementation of ECB (Electronic Codebook) encryption mode with key size 128 before moving on to other modes as time permits. The performance of our accelerator will be assessed based on power efficiency for area used and completing the encryption/decryption in fewer cycles than a general purpose CPU.

1 INTRODUCTION

The purpose of this project surrounds the increasing popularity of wireless wearable technology. Devices such as smartwatches or smart rings have disrupted traditional healthcare, as wearable devices can be equipped with sensors capable of continuous health monitoring and fitness tracking [7]. With an estimated 534m global wearable unit shipments in 2024 [3], nearly everyone has some form of a wearable device. Many of these devices have unparalleled access to your life, collecting and transmitting data such as location, audio, biometrics, or health data, making privacy and security a major concern. Thus, each of these devices require cryptography to ensure the data is not vulnerable to cyberattacks, interceptions, and other unauthorized access.

Unfortunately cryptography within wearable devices is not simple. Encryption on these small devices require a delicate balance between security and power efficiency. The more complex algorithms which are more secure are also computationally expensive which makes them an ill choice for battery powered devices [7]. The status quo solution has long been AES encryption through software on a general purpose CPU, as it offers a good balance of security and performance. However, encryption on specialized hardware using an AES accelerator allows for higher performance in terms of cycles to compute and by extension lower energy usage due to less cycles needed to encrypt and decrypt information [5]. However, strong reason companies tend to lean towards a software implementation is cost. Added silicon from an AES accelerator is not free, developing their own solution and research is costly, while proprietary chips with built in AES accelerators such as Samsung's Exynos SOC can be expensive.

We aim to develop a system-on-chip that allows AES accelerators to be a better solution than the software based solution within wearable devices. By utilizing a specialized ASIC we can reduce the space required for our accelerator, and design our SOC to be power efficient [6]. Furthermore, through utilizing open source software and hardware we can reduce time and money spent, making our solution more cost effective. With less area, power usage and an open source solution, it should reduce the costs enough to justify using a hardware accelerator. Additionally, modern wearable devices can process thousands if not millions of data points per minute, which will require a high throughput of data to be encrypted. To meet the throughput requirements we will utilize an AXI protocol to facilitate communication between the CPU core and our AES accelerator module.

Furthermore, our solution will be completed with minimal changes to the CPU core itself, which will allow wearable technology designers to have an easier time adopting our proposed ASIC. We will deliver a modern solution to ensure the security of lower-end wearable devices, and allow consumers a peace of mind knowing their data is safely encrypted.

2 PROPOSED APPROACH

We propose to design a low-power AES-128 hardware accelerator integrated with PicoRV32 RISC-V processor through AXI4 memory-mapped interface, targeting secure wearable devices. The AES accelerator will handle cryptographic operations such as data encryption and decryption directly in hardware, offloading the CPU and significantly reducing power and latency compared to software-based AES. A shared on-chip memory will facilitate data exchange between the CPU and accelerator without the need for software protocol changes.

Wearable devices demand low-power consumption, compact area and hardware level security – all of which are best achieved through an ASIC implementation. Using TSMC 16 nm technology enables high performance at low voltage, ensuring energy-efficient encryption suitable for battery-powered wearable systems.

The target audience includes wearable device designers, IoT security engineers, and SoC developers focused on healthcare and fitness tracking applications. This work benefits the wearable industry by providing a secure, lightweight, and energy-efficient AES core that can be embedded in RISC-V-based SoCs for data protection in connected devices.

While most existing AES accelerator integrations target high-performance, 5-stage, or out-of-order processors to maximize throughput, our work focuses on energy efficiency and simplicity for wearable devices. We integrate a lightweight, single-stage PicoRV32 RISC-V core with a hardware AES-128 accelerator through an AXI4 memory-mapped interface and a shared on-chip SRAM, allowing direct data access by both units.

This design significantly reduces area, power, and verification complexity, making it ideal for battery-operated systems. However, the trade-off is a lower instruction throughput and reduced CPU parallelism compared to multi-stage or out-of-order architectures. Despite this, system-level encryption throughput remains high since the AES accelerator handles the computational workload autonomously.

Overall, the work demonstrates a balanced power–area–performance architecture optimized for secure wearable applications.

3 DESIGN CONSTRAINT

Although our constraints are for wearable technology as a whole, we will use a Samsung Galaxy Watch 7 for comparison because information regarding its SOC is readily available.

With a focus on being cost effective, we seek to minimize area usage to reduce silicon costs. As such, our constraints fall on using comparable silicon to the current status-quo, for instance, the SOC used in Samsung devices is their Exynos W1000 (3nm) utilizing 17.67 mm^2 . Furthermore, for our use on wearable devices which are usually battery powered, we set strict constraints on our power usage.

The Galaxy Watch 7 has a battery of 325mAh with a 5V output. The battery life of the device is about a day (24 hours) which calculates to a power consumption of 83 mW.

We can set some loose constraints based on calculations from the Galaxy Watch 7.

- Chip Area: 17.67 mm^2
- Average Power: 83 mW

For our specific metrics, we will be using one core compared to the Exynos's 5 cores, so we can consume much less area. The only other main modules on the chip will be our AES and SRAM.

Much of the batteries of the watch are consumed by peripherals (such as sensors or screen) which are not in every wearable device, so we aim to have much lower power consumption.

- Chip Area: $4.5mm^2$
 - : $3.8mm^2$ (RISC-V CPU)
 - : $0.2mm^2$ (AES Accelerator)
 - : $0.5mm^2$ (SRAM)
- Average Power: $10mW$

4 RELATED WORK

Recent studies have focused on enhancing AES performance and efficiency for embedded and IoT systems through hardware acceleration and RISC-V customization. The authors in [5] presented a RISC-V core with an integrated AES-256 accelerator connected via an AXI4-Lite memory-mapped interface. Their implementation achieved up to 84% faster encryption and significant power savings compared to pure software designs. Work in [1] developed AES-RV, a hardware-efficient RISC-V accelerator using low-latency custom AES instruction extensions. They achieved higher energy and area efficiency, highlighting the benefits of tightly optimized pipelines and custom instructions. In a similar work [4], RVCP, a RISC-V processor is used as a co-processor with a specialized cryptographic accelerator unit tightly coupled to the processor datapath. Their implementation achieved a higher hardware efficiency compared to CPU implementations.

Building on these prior works, we propose a low-area, low-power AES accelerator that is loosely coupled to a RISC-V host via memory-mapped I/O, specifically optimized for wearable and energy-constrained systems.

5 SYSTEM-ON-CHIP ARCHITECTURE

Core Components

- RISC V CPU core - PicoRV32
- AES accelerator - 128 key size and Electronic Code Book(ECB) AES mode

Enhancements

- Configurable key size
- Other AES modes
 - Cipher Block Chaining (CBC)
 - Cipher Feedback (CFB)
 - Output Feedback (OFB)
 - Counter (CTR)

The core components in our design are the functional components that are required to perform basic computations and basic AES operations. The enhancements will be the additional functionalities that we plan to implement on the AES accelerator if the time permits.

The RISC-V processor is an open-source design PicoRV32[2] which is implemented in RTL. The processor will act as the main computation unit in the SoC and will handle the generic computations. We will be removing the unnecessary components and interfaces in the RISC-V processor which are not related to our application to match the constraints.

The AES accelerator module, which handles the AES workloads will be developed from scratch in RTL. As the base implementation, we will be implementing the AES encryption and decryption with key size of 128bit and ECB AES mode. Main components in this module are the controller, the AES data-path and the key scheduler. We will be implementing other AES modes and the key

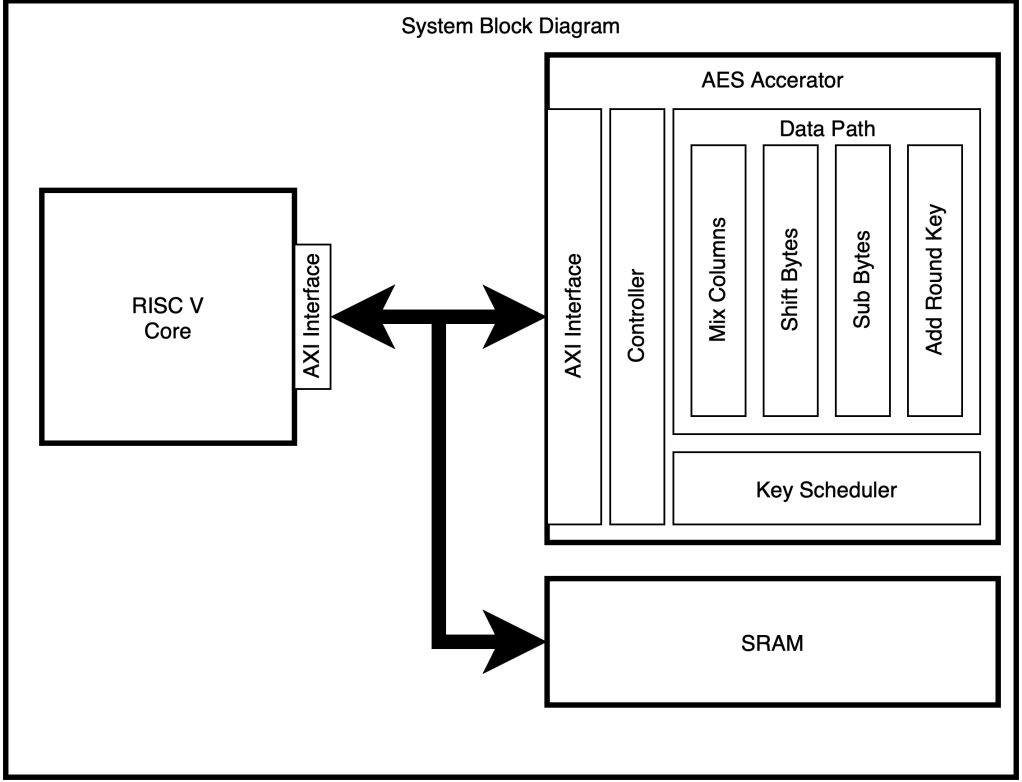


Fig. 1. ASIC block diagram. Primary components: RISC-V processor, AES Accelerator; SRAM

size will be made configurable (128, 196, 256) as enhancements to the base implementation. The AES accelerator will be integrated as a loosely coupled accelerator with memory mapped I/O. An AXI interconnect which we will be implementing from scratch in RTL will be used as the main communication bus in the design. The RISC V processor and the AES accelerator will be accessing the memory via the AXI interconnect.

6 PROPOSED MILESTONES

Tasks and milestones for our project are described in Fig. 2.

7 VERIFICATION

We propose a multi-layered verification approach that provides comprehensive coverage from algorithmic correctness through physical implementation, ensuring a robust, tape-out-ready design.

7.1 RTL Functional Verification

The AES accelerator and integrated SoC will undergo extensive RTL simulation using comprehensive test-benches. Functional correctness of the AES-128 core will be verified against NIST standard test vectors (FIPS 197) to validate both encryption and decryption operations. Directed and constrained-random test-benches will exercise various operational scenarios, including back-to-back transactions, pipeline stalls, and error conditions. Code coverage analysis will be performed

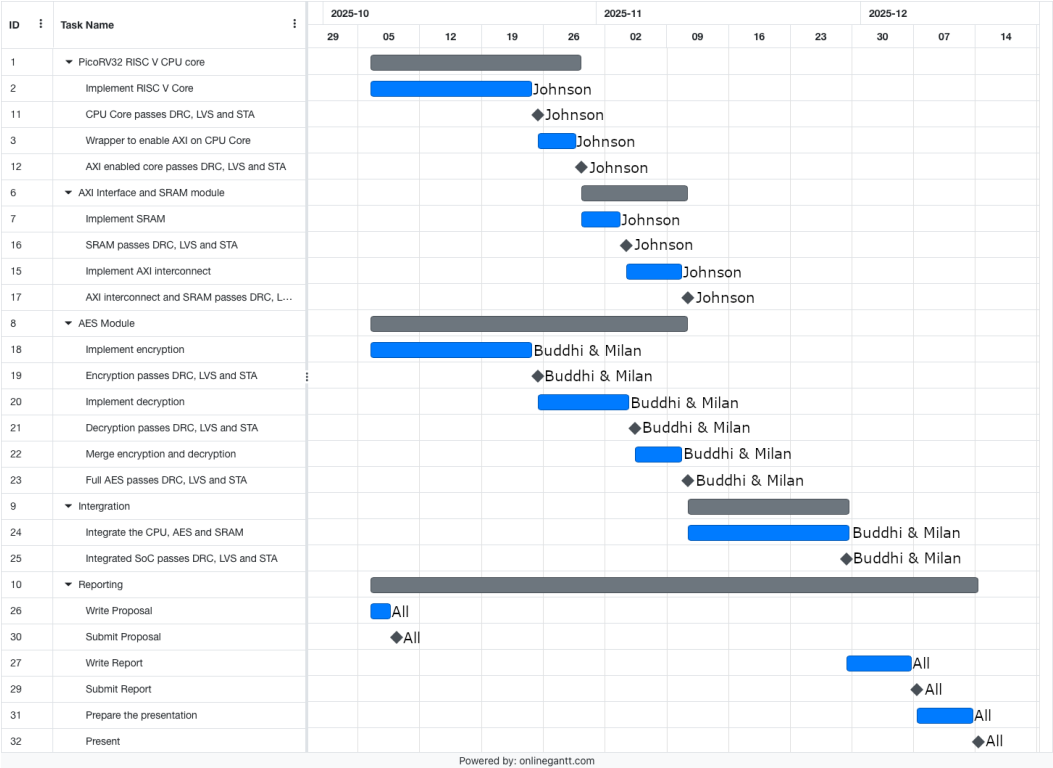


Fig. 2. Gantt Chart of SOC Development

using Cadence IMC (Incisive Metric Center) to ensure thorough verification coverage. Functional coverage metrics will track key design scenarios including all AES operation modes, register access patterns, and boundary conditions. The goal is to achieve >95% code coverage and 100% coverage of critical functional scenarios.

7.2 Physical Verification

Following place-and-route, the design will undergo mandatory physical verification:

- DRC(Design Rule Check): Using industry-standard tools to verify compliance with TSMC 16nm PDK design rules
- LVS(Layout Versus Schematic: To ensure the physical layout matches the netlist, verifying all connections and device parameters

7.3 Timing Verification

Static Timing Analysis (STA) will be performed across all PVT (Process, Voltage, Temperature) corners using Cadence Tempus.

7.4 Power Analysis

Post-layout power analysis will estimate static and dynamic power consumption using switching activity from simulation. IR drop analysis will verify power grid integrity and ensure voltage droop remains within acceptable limits.

Table 1. Required IP for this project, including tools and designs.

Task	Planned Solution
RTL Simulation	Synopsys VCS
Code Coverage	Cadence IMC Tool
Logic Synthesis	Cadence Genus
Place and Route	Cadence Innovus
STA	Cadence Tempus
DRC	Cadence Innovus
LVS	Mentor Calibre
Timing Verification	Cadence Tempus
CPU Core	PicoRV32
PDK	TSMC N16 ADFP

8 EVALUATION

We will evaluate our success based on functionality and how well we meet our constraints. Functionality would be measured based on the correctness of encryptions and decryptions; the pre-encryption message should be the same as the post-decryption message. This would ensure the module is operating as expected. Also, importantly, we want to ensure we are meeting our constraints. This means our chip area is less than 4.302 mm^2 , and power consumption is within 1mW. We will also include a comparison to the status-quo baseline solution, which is running AES on a general purpose CPU. This comparison will be done through comparing the cycles our accelerator takes versus the cycles an OpenSSL software implementation will take for AES operations [5]. We can deem our design successful if we can achieve a functional design with comparable area and power to state-of-the-art solutions while taking fewer cycles than the status-quo solution.

9 REQUIRED MATERIALS

Table 1 reflects the materials and tools the team will need to complete the project. For most tasks we will use commercial software such as Cadence or Synopsys, but we will explore open source solutions when possible. For instance, we will be using an open-source RISC-V CPU core as our processor.

10 WORKLOAD DISTRIBUTION

10.1 Backgrounds

10.1.1 Buddhi. He has a background in computer architecture and hardware design. He has taken graduate-level classes at NYU in VLSI Design (ECE 6473/6443) and Computing Systems Architecture (ECE 6913). He has industry experience as a computer hardware engineer and has worked on the verification of RISC-V CPU designs. Buddhi’s goal is to gain experience on the RTL-to-GDS flow and to gain experience in accelerator design.

10.1.2 Milan. He has a background in VLSI design and FPGA design. He has taken ECE 6473 (Intro to VLSI Design), ECE 6443 (VLSI Design and System Architecture) and ECE 6913 (Computing System Architecture) at NYU. He has done a few projects in hardware design. His goal is to gain experience in RTL Design and verification.

10.1.3 Johnson. He has studied CPU architecture (ECE 6913) and VLSI design (ECE 6443) from his graduate studies. Furthermore, he has completed personal projects such as a software implementation of a 5-stage CPU pipeline. His goal is to gain better understanding of the process of fabricating a chip from design to tapeout, and gain familiarity with the tools to do so.

10.2 Distribution

The Gantt chart described in Fig. 2. reflects our tentative distribution. Each member will be responsible for their respective task, but are expected to be familiar with each other's work. Our distribution leans more heavily on the AES accelerator; it will consume the bulk of our work and time so we have allotted more manpower to that task. Our main blocks are as follows:

- AES: Buddhi and Milan
- CPU & SRAM: Johnson

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